

《电磁场与电磁波》所用的复杂公式（会附在考试试卷后）

	$\vec{e}_x$	$\vec{e}_y$	$\vec{e}_z$
$\vec{e}_\rho$	$\cos\phi$	$\sin\phi$	0
$\vec{e}_\phi$	$-\sin\phi$	$\cos\phi$	0
$\vec{e}_z$	0	0	1

	$\vec{e}_x$	$\vec{e}_y$	$\vec{e}_z$
$\vec{e}_r$	$\sin\theta\cos\phi$	$\sin\theta\sin\phi$	$\cos\theta$
$\vec{e}_\theta$	$\cos\theta\cos\phi$	$\cos\theta\sin\phi$	$-\sin\theta$
$\vec{e}_\phi$	$-\sin\phi$	$\cos\phi$	0

圆柱坐标系下：

$$\nabla u = (\hat{e}_r \frac{\partial u}{\partial \rho} + \hat{e}_\phi \frac{1}{\rho} \frac{\partial u}{\partial \phi} + \hat{e}_z \frac{\partial u}{\partial z}) \quad \nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \frac{\partial u}{\partial \rho}) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

$$\nabla \cdot \vec{F} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \quad \nabla \times \vec{F} = \frac{1}{\rho} \begin{vmatrix} \hat{e}_\rho & \rho \hat{e}_\phi & \hat{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_\rho & \rho F_\phi & F_z \end{vmatrix}$$

球坐标系下： $\nabla u = (\hat{e}_r \frac{\partial u}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial u}{\partial \theta} + \hat{e}_\phi (\frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi}))$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial u}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial u}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

$$\nabla \cdot \vec{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$$

$$\nabla \times \vec{F} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{e}_r & r \hat{e}_\theta & r \sin \theta \hat{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & r F_\theta & r \sin \theta F_\phi \end{vmatrix}$$

电磁场 一般边界条件	两种媒质均为非理 想介质的分界面上	两种理想介质分界面 上的边界条件	理想导体表面上的边 界条件
$\begin{cases} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \end{cases}$	$\begin{cases} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = 0 \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \end{cases}$	$\begin{cases} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = 0 \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = 0 \end{cases}$	$\begin{cases} \vec{e}_n \times \vec{H}_1 = \vec{J}_s \\ \vec{e}_n \times \vec{E}_1 = 0 \\ \vec{e}_n \cdot \vec{B}_1 = 0 \\ \vec{e}_n \cdot \vec{D}_1 = \rho_s \end{cases}$

静电位边界条件： $\begin{cases} \varphi_1 = \varphi_2 \\ \varepsilon_2 \frac{\partial \varphi_2}{\partial n} - \varepsilon_1 \frac{\partial \varphi_1}{\partial n} = \rho_s \end{cases}$ 。 积分 $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2}) + C$

3.1 节：电偶极子的电位 $\varphi(\vec{r}) = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^3}$,

电偶极子的远区电场强度 $\vec{E}(\vec{r}) = \frac{p}{4\pi\epsilon_0 r^3}(\hat{e}_r 2\cos\theta + \hat{e}_\theta \sin\theta)$

3.3 节：小圆环电流的矢量磁位 $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi r^3} \vec{p}_m \times \vec{r}$

小圆环电流的远区磁感应强度 $\vec{B}(\vec{r}) = \frac{\mu_0 IS}{4\pi r^3}(\hat{e}_r 2\cos\theta + \hat{e}_\theta \sin\theta)$

4.2 节：洛伦兹规范条件下，矢量位和标量位满足的微分方程（达朗贝尔方程）

$$\nabla^2 \vec{A} - \mu\epsilon \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu \vec{J}; \nabla^2 \varphi - \mu\epsilon \frac{\partial^2 \varphi}{\partial t^2} = -\frac{1}{\epsilon} \rho$$

4.3 节：电磁能量守恒定律 $-\frac{d}{dt} \int_V w dV = \int_V \vec{J} \cdot \vec{E} dV + \oint_S \vec{S} \cdot d\vec{S}$

5.2.1 节：导电媒质中的均匀平面波（一般情况下）的

$$\text{衰减常数 } \alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} - 1 \right]} \quad \text{相位常数 } \beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon} \right)^2} + 1 \right]}$$

5.2.2 节：弱导电媒质的本征阻抗近似为 $\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon} \right)$

5.2.3 节：良导体的本征阻抗近似为 $\eta_c \approx \sqrt{\frac{2\pi f \mu}{\sigma}} e^{j\pi/4}$

第七章：导行电磁波用纵向场分量表示横向场分量（7.1.5）式：

$$H_x = -\frac{1}{k_c^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right); H_y = -\frac{1}{k_c^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$

$$E_x = -\frac{1}{k_c^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right); E_y = -\frac{1}{k_c^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$

式中 $k_c^2 = \gamma^2 + k^2$, k 为无界均匀媒质中电磁波的波数。

（创新班）第六章：对理想介质分界面的垂直入射反射波电场振幅、透射波电场振幅

$$E_{rm} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} E_{im}; E_{tm} = \frac{2\eta_2}{\eta_2 + \eta_1} E_{im} \quad 6.2 \text{ 节 等效波阻抗 } \eta_{ef} = \eta_2 \frac{\eta_3 + j\eta_2 \tan(\beta_2 d)}{\eta_2 + j\eta_3 \tan(\beta_2 d)}$$